AP CALCULUS AB

Unit 1 • Unit 1 Pretest • Limits, Continuity, and Differentiability

# Unit 1 Pretest • Limits, Continuity, and Differentiability

Name: _____ Date: _____

**Directions**

Answer every question. Multiple-choice questions have one best answer. On free-response questions, show your work and justify your answers. Questions labeled **NO CALCULATOR** must be done without a calculator; questions labeled **CALCULATOR** require a graphing calculator. For questions with a graph, read the values you need from the grid.

## Part A • Multiple choice

### Question 1

NO CALCULATOR

$\lim_{x \to 3} \dfrac{x^2 - 9}{x - 3}$ is

(A) 0

(B) 3

(C) 6

(D) nonexistent

### Question 2

NO CALCULATOR

$\lim_{x \to \infty} \dfrac{5x^2 - x}{2x^2 + 3}$ is

(A) 0

(B) $\frac{5}{2}$

(C) 5

(D) $\infty$

### Question 3

NO CALCULATOR

$\lim_{x \to 1^-} \dfrac{2}{x - 1}$ is

(A) $\infty$

(B) 2

(C) 0

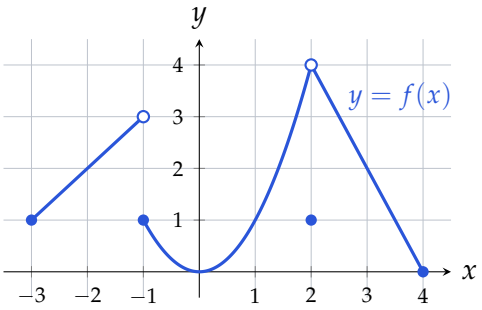
(D) $-\infty$

### Question 4

NO CALCULATOR

$\lim_{x \to 2} \dfrac{x^2 + x - 6}{x - 2}$ is

(A) 0
(B) 1
(C) 5
(D) nonexistent




### Question 5

NO CALCULATOR

Using the graph of $f$, which limit does not exist?

(A) $\lim_{x \to 0} f(x)$
(B) $\lim_{x \to -1} f(x)$
(C) $\lim_{x \to 2} f(x)$
(D) $\lim_{x \to 3} f(x)$

### Question 6

NO CALCULATOR

Using the graph of $f$, at which $x$ is the discontinuity removable?

(A) $x = -1$
(B) $x = 0$
(C) $x = 2$
(D) $x = 3$

### Question 7

NO CALCULATOR

For which $k$ is $f(x) = \begin{cases} kx^2, & x \leq 1 \\ x + 3, & x > 1 \end{cases}$ continuous?

(A) 4
(B) 3
(C) 1
(D) 2

### Question 8

NO CALCULATOR

$f$ is continuous on $[2,6]$, $f(2)=5$, $f(6)=-3$. Which must be true?

(A) $f(4)=1$
(B) $f(c)=0$ for some $c$
(C) $f$ has no maximum
(D) $f'(c)=-2$

### Question 9

NO CALCULATOR

$f$ is differentiable, $f(1)=2$ and $f(5)=10$. Which must be true?

(A) $f'(c)=2$ for some $c$ in $(1,5)$
(B) $f'(c)=8$ for some $c$ in $(1,5)$
(C) $f(3)=6$
(D) $f'(c)=0$ for some $c$ in $(1,5)$

### Question 10

NO CALCULATOR

$f(x)=|x+1|$ is not differentiable at

(A) $x=1$
(B) $x=0$
(C) $x=-1$
(D) nowhere

## Part B • Free response

### Question 11 (Free response)

NO CALCULATOR

Let $g(x)=\begin{cases}\dfrac{x^2+x-6}{x-2}, & x<2\\ cx+1, & x\geq 2.\end{cases}$

(a) Find $\lim_{x\to 2^-} g(x)$.
(b) Find $c$ so that $g$ is continuous at $x=2$; justify with the definition.
(c) With this $c$, is $g$ differentiable at $x=2$? (The left piece simplifies to $x+3$.)

### Question 12 (Free response)

NO CALCULATOR

A differentiable function $W$ (temperature, °C) has $W(0)=20$, $W(3)=24$, $W(5)=31$, $W(8)=27$.

(a) Must $W(t)=30$ for some $t$ in $(3,5)$? Justify.
(b) Must $W'(t)=3.5$ for some $t$ in $(3,5)$? Justify.
(c) Estimate $W'(4)$, with units.